



Bachelor Project (2022)

**Applications of Van der Waerden's Theorem
in the ring of polynomials over $\mathbb{F}_p$**

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Abstract

In this paper, we prove van der Waerden's Theorem in the context of polynomials over the finite ring $\mathbb{Z}/n\mathbb{Z}$, showing that any finite colouring of this ring of polynomials admits arbitrarily long generalised arithmetic progressions. We then use this to show the result that one can find two polynomials p, q such that $p, pq, p + q$ are all the same colour, originally proven by Moreira in $\mathbb{N}$. The notion of summation in this ring is similar to that of the bitwise XOR function in binary, and we furthermore show that it is possible to find a monochromatic quadruplet $x, y, x +_n y, x \oplus_m y$ where x, y are strings and $+_n$ is their summation in standard base- n arithmetic; $\oplus_m$ is their summation in m -modular bitwise arithmetic. We finally give a corollary to colouring subsets of $\mathbb{N}$ and show some interesting results in this new structure.

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Introduction

In 1927, Bartel Leendert van der Waerden published one of the most notable results of additive number theory, now named after him¹:

Theorem 1: Van der Waerden's Theorem

For any positive integers r and k , there exists N such that if every element in the set $\{1, 2, \dots, N\}$ is coloured in one of r colours, there are at least k monochromatic integers in arithmetic progression.

This theorem can also be expressed in the form of a corollary about colouring all of $\mathbb{N}$:

Corollary 2: VdW; the infinite case

For any finite colouring of $\mathbb{N}$, we can find arbitrarily long monochromatic arithmetic progressions.

This was a significant strengthening of David Hilbert's cube Lemma from 1894. Van der Waerden considered this to be an isolated problem and so it would be two more decades before Paul Erdős would introduce what we now call Ramsey Theory, built on the foundations of Van der Waerden's theorem, along with results from Issai Schur and the eponymous Frank Ramsey.

Whilst Ramsey Theory has a relative wealth of results concerning completely additive structures, and these results can be translated relatively easily to a multiplicative format (simply by considering powers of 2) it is quite difficult to combine the two. For instance, given a finite colouring of $\mathbb{N}$, it remains an open conjecture if we can find a monochromatic quadruplet of the form $\{x, y, x + y, xy\}$. In 2017, Joel Moreira proved an intermediary result about the existence of $\{x, x + y, xy\}$, more than 90 years after Schur proved the existence of $\{x, y, x + y\}$ and $\{x, y, xy\}$.

The aim of this paper is to investigate how these results translate to a new environment - the polynomial ring $\mathbb{Z}/n\mathbb{Z}[t]$.

¹van der Waerden, B. L. (1927). *Beweis einer Baudetschen Vermutung*. Nieuw. Arch. Wisk. (in German). 15: 212–216

A polynomial party

Every journey needs a map, so let us set down some definitions and lay down the ground-work.

1.1 Setting the scene

Firstly, we must lay our foundations. We leave the relative comforts and familiarity of $\mathbb{N}$ for newer grounds: the polynomial ring over a finite field.

Definition 1.1 We refer to the ring of polynomials with coefficients in $\mathbb{Z}/n\mathbb{Z}$ as $\mathbb{Z}/n\mathbb{Z}[t]$, with addition and multiplication defined as usual.

Definition 1.2 We refer to the subset of polynomials in $\mathbb{Z}/n\mathbb{Z}[t]$ with degree at most k as $\mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$.

Definition 1.3 Let $\Phi_k = \{\phi_0, \phi_1, \phi_2, \dots, \phi_{k-1}\} \subset \text{End}(\mathbb{Z}/n\mathbb{Z}[t])$ henceforth represent a subset (of cardinality $k \geq 2$) of endomorphisms^a of $\mathbb{Z}/n\mathbb{Z}[t]$. Such a subset is called rooted if ϕ_0 is the neutral element of $\text{End}(\mathbb{Z}/n\mathbb{Z}[t])$, namely the constant zero endomorphism. A Φ_k -AP (arithmetic progression) is a subset $\{p + \phi(q) \mid \phi \in \text{rooted } \Phi_k\} \in \mathbb{Z}/n\mathbb{Z}[t]$ containing a polynomial all the shifts of a polynomial p by the various images of a polynomial q (the difference) by the elements of a **rooted** Φ . If Φ_k is rooted, we say it is a rooted Φ_k -AP. More generally, we use $p + \Phi(q)$ as a shorthand for the subset $\{p + \phi(q) \mid \phi \in \Phi_k\} \in \mathbb{Z}/n\mathbb{Z}[t]$.

^aHere, we refer to a group homomorphism rather than a ring homomorphism; no multiplicative property is required, but simply that $\phi(p + q) = \phi(p) + \phi(q)$.

An interesting thing to note is that an endomorphism in $\mathbb{Z}/n\mathbb{Z}[t]$ is uniquely determined by the images of $1, t, t^2, \dots, t^n, \dots$ as the image of an arbitrary polynomial follows from the additive structure that homomorphisms are endowed with by definition.

Definition 1.4 A family of sets $\mathcal{P} \subseteq \mathcal{P}(\mathbb{Z}/n\mathbb{Z}[t])$ is said to be upward closed if for all $A \subseteq B \subseteq \mathcal{P}(\mathbb{Z}/n\mathbb{Z}[t])$, $A \in \mathcal{P} \implies B \in \mathcal{P}$. The dual family of $\mathcal{P}$, which we denote by $\mathcal{P}^*$, is defined as

$$\mathcal{P}^* = \{A \subset \mathbb{Z}/n\mathbb{Z}[t] \mid A \cap B \neq \emptyset, \forall B \in \mathcal{P}\}$$

Definition 1.5 A family of sets $\mathcal{P} \subseteq \mathcal{P}(\mathbb{Z}/n\mathbb{Z}[t])$ is said to be partition regular if for any finite colouring of a set $A \in \mathcal{P}$ there exists a monochromatic subset of A that belongs to $\mathcal{P}$.

A classic example of a partition regular family is the sets with infinite cardinality. If you apply a finite colouring to an infinite set, one of the resultant subsets must clearly be infinite. We can now prove a couple of interesting results about upward closed sets and their dual:

Property 1.6 Let $\mathcal{P}$ be an upward closed family. Then the following is true:

1. For any set $A \subseteq X$ we have

$$A \in \mathcal{P} \iff A^c \notin \mathcal{P}^*.$$

2. $\mathcal{P}^{**} = \mathcal{P}$.

Proof. The direction $\implies$ is pretty clear in 1; this follows more or less immediately from our definition of a dual. For the other direction, suppose A^c is not in $\mathcal{P}^*$; this implies some set $B \in \mathcal{P}$ did not intersect with A^c , but this implies in turn that $B \subseteq A$ and so by upward closure, $A \in \mathcal{P}$.

For 2, we simply apply the previous result: $A \in \mathcal{P} \iff A^c \in \mathcal{P}^*$; $A^c \in \mathcal{P}^* \iff A \in \mathcal{P}^{**}$, proving $\mathcal{P} = \mathcal{P}^{**}$ as desired.

Property 1.7 Let $\mathcal{P}$ be an upward closed family. Then we have that the family $\mathcal{P} \wedge \mathcal{P}^* := \{A \cap B \mid A \in \mathcal{P}, B \in \mathcal{P}^*\}$ is upward closed and partition regular.

Proof. Upward closure is more or less immediate from the definition.

A key idea when proving partition regularity is that it suffices to provide a proof for the case with two colours, as any finite number of colours follows by induction. We therefore consider $C \in \mathcal{P} \wedge \mathcal{P}^*$ and split it into two disjoint sets C_1 and C_2 with $C = C_1 \cup C_2$. We only need to prove that a monochromatic subset is in $\mathcal{P} \wedge \mathcal{P}^*$ but as we have upward closure, we know we can prove the marginally stronger result that either C_1 or C_2 is in $\mathcal{P} \wedge \mathcal{P}^*$. By definition, $\exists A \in \mathcal{P}, B \in \mathcal{P}^*$ such that $C = A \cap B$. Let $D := C_1 \cup A^c$. By the previous property, we know either $D \in \mathcal{P}^*$ or $D^c \in \mathcal{P}$, but as $C_1 = A \cap D$ and $C_2 = D^c \cap B$, in either case we have the desired result.

1.2 Preliminary results: syndetic and thick sets

Two very natural definitions in the domain of combinatorial number theory in $\mathbb{N}$ are those of syndetic sets and thick sets; the former being sets with bounded "gaps" (a gap being the distance between two consecutive elements of a set) and the latter being sets with arbitrarily long "intervals" (an interval being a subset consisting of elements that are consecutive in $\mathbb{N}$). These two concepts do not seem to translate naturally to our new setting, however, where we now no longer have a natural notion of complete ordering or distance. However, we can use an alternative definition of syndetic sets in $\mathbb{N}$ - sets which cover $\mathbb{N}$ with a finite number of shifts² - as an inspiration for how to proceed in $\mathbb{Z}/n\mathbb{Z}[t]$.

Definition 1.8 A syndetic set in $\mathbb{Z}/n\mathbb{Z}[t]$ is a subset $S \subseteq \mathbb{Z}/n\mathbb{Z}[t]$ such that there exists a finite set $F \subset \mathbb{Z}/n\mathbb{Z}[t]$, $\mathbf{0} \in F$ such that the sumset $S + F = \mathbb{Z}/n\mathbb{Z}[t]$ - alternatively, we can say that every polynomial in $\mathbb{Z}/n\mathbb{Z}[t]$ can be written as the sum of an element in S and an element in F .

Let us see if we can find a way to classify all the syndetic sets. Firstly, as F is finite, it contains an element a of maximal degree amongst all elements of F (this maximal element need not be unique, of course). We henceforth refer to the $\deg(a) = k$ as the *signature* of the syndetic set in question. We can therefore extend F to $\mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$, the set of all polynomials of degree at most k , which is a finite set as there are clearly p^k such polynomials - we just have to choose the value of each coordinate. We can therefore assume we are working with the larger set $\mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$, as the condition that $S + \mathbb{Z}/n\mathbb{Z}[t]_{\leq k} = \mathbb{Z}/n\mathbb{Z}[t]$ still holds.

Now consider the equivalence relation $b \sim c \iff b - c \in \mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$, and the induced equivalency classes $\mathbb{Z}/n\mathbb{Z}[t] / \mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$. We have that S must contain at least one element from every equivalency class as otherwise none of the elements in that equivalency class will be in the sumset. Furthermore this is also sufficient, since every other element of the equivalency class can be reached.

An equivalency class is simply a set of valid³ coefficients for $t^{k+1}, t^{k+2}, \dots$ as two polynomials that are equal over these coefficients have a difference in $\mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$, and if two polynomials differ in the coefficient of $t^m, m > k$ then their difference will have degree $\geq m$ and therefore not be in $\mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$. Therefore, a syndetic set S is any set containing a polynomial for every possible set of valid coefficients for $t^x, x > k$, where k is the signature of our syndetic set.

A very natural question to ask is how to define thick sets, as there is no alternative definition which works nicely in $\mathbb{Z}/n\mathbb{Z}[t]$. The trick is using the property that the family of thick sets is dual to the family of syndetic sets, and turning this from a property to the basis of the definition.

²There exists F finite such that $\mathbb{N} = \bigcup_{f \in F} (S - f)$, where $(S - f) := \{s - f \mid \forall s \in S\} \cap \mathbb{N}$.

³In the sense that the polynomial still has a maximal degree - from some point on, all coefficients are 0.

Definition 1.9 A thick set in $\mathbb{Z}/n\mathbb{Z}[t]$ is a subset $T \subseteq \mathbb{Z}/n\mathbb{Z}[t]$ such that for every syndetic set $S \subseteq \mathbb{Z}/n\mathbb{Z}[t]$, $S \cap T \neq \emptyset$.

Now, by definition, the thick sets will be dual to the syndetic sets, so we don't have to prove this anymore. What we do need to do, however, is understand the nature of thick sets. As we have seen, a syndetic set must contain an element for any given set of coefficients from some value onwards - the signature of the syndetic set. For example, a syndetic set with signature 3 with containing a polynomial for any set of coefficients from the third degree onwards. This means it must contain some element of the form $2t^4 + t^3 + \dots$ or $3t^4 + t^3 + \dots$ or $t^5 + t^4 + 2t^3 + \dots$ or any other combination under the sun. We cannot however say anything about the lower coefficients - it might contain no polynomials with $2t^2$, or 3, or so forth. This is analogous to the notion of equivalency classes from before - a set of coefficients from some degree onwards *is* an equivalency class. We will therefore also refer to the signature of an equivalency class in a similar manner.

What must a thick set therefore contain? Suppose the thick set has the polynomial $2t^4 + t^3 + 2t^2 + t + 3$. It would therefore intersect with all the syndetic sets of signature 3 that contain this polynomial as one of their representatives for the equivalency class $[2t^4 + t^3]_{\sim}$. However, there may be others instead containing another member of the equivalency class, such as $2t^4 + t^3 + 3t + 1$. As long as there is *some* polynomial in the equivalency class $[2t^4 + t^3]_{\sim}$ not in our thick set, we are potentially not intersecting a syndetic set. Of course, one might argue we might intersect these syndetic sets elsewhere, but as long as our thick set does not contain an entire equivalency class, we can just construct a disjoint syndetic set by selecting the elements in each equivalency class not in our thick set. Therefore, to intersect with all the syndetic sets of signature 3, our thick set must contain an equivalency class of signature 3 in its entirety.

Since our thick set must intersect with all syndetic sets, and not just those of signature 3, so we must contain equivalency classes for any signature $k \in \mathbb{N}$. Containing an equivalency class of signature N means you also contain one of signature $N - 1, N - 2, \dots$ and so we can classify a thick set as one that contains equivalency classes of arbitrarily large signature.

This actually gives us some insight into how the notion of a gap or an interval in $\mathbb{N}$ translates to $\mathbb{Z}/n\mathbb{Z}[t]$ - whilst syndetic sets in $\mathbb{N}$ were those that contained elements in any interval of a given size, and thick sets contained intervals of any size, we have now replaced intervals with equivalency classes. There is a very natural transformation from $\mathbb{N}$ to $\mathbb{Z}/n\mathbb{Z}[t]$ - writing a number in base n and using the representation for the coefficients, and in the opposite direction, evaluating a polynomial in n . This bijection has a lot of desirable properties, most notably that it gives a well-ordering of $\mathbb{Z}/n\mathbb{Z}[t]$. A natural question to ask is if it preserves thickness and syndeticity.

Property 1.10 Let $\mathfrak{E}_n$ denote the forward direction of bijection between $\mathbb{N}$ and $\mathbb{Z}/n\mathbb{Z}[t]$ described above. Then we have that a set S is thick (respectively syndetic) in $\mathbb{N}$ if and only if $\mathfrak{E}_n(S)$ is thick (respectively syndetic) in $\mathbb{Z}/n\mathbb{Z}[t]$.

Proof. Consider a thick set $T \in \mathbb{N}$; this must contain arbitrarily long intervals. In particular, it contains an interval of length at least $2 \cdot n^k$, for any k . This is a set of integers that, when written in base n , must contain (by the pigeonhole principle) all the possible permutations of the smallest k digits, for a fixed choice of the greater digits. The image of this subset will be a equivalency class of signature k ; as this holds for arbitrarily large k , $\mathfrak{E}_n(T)$ is thick. In the reverse direction, if T is thick in $\mathbb{Z}/n\mathbb{Z}[t]$ it contains arbitrarily large equivalency classes, which are arbitrarily large intervals in $\mathbb{N}$.

Consider a syndetic set $S \in \mathbb{N}$; this has gaps bounded by some $h \in \mathbb{N}$ and so in particular for $p^k > h$, the set S contains at least one element which can be written as $ap^k + b$, $0 \leq b \leq p^k - 1$ for every $a \in \mathbb{N}$. When taken in $\mathbb{Z}/n\mathbb{Z}[t]$, these elements correspond to an element from every equivalency class as described after Definition 6. In the reverse direction, if S is syndetic in $\mathbb{Z}/n\mathbb{Z}[t]$ with signature p^k then $\mathfrak{E}_n^{-1}(S)$ has gaps bounded by $2p^k$.

1.3 Piecewise syndetic sets

Property 1.11 *A piecewise syndetic set in $\mathbb{Z}/n\mathbb{Z}[t]$ is a subset $P \subseteq \mathbb{Z}/n\mathbb{Z}[t]$ such that there exist a thick set T and a syndetic set S with $P = S \cap T$.*

Property 1.12 *Let $\mathfrak{E}_n$ denote the forward direction of bijection between $\mathbb{N}$ and $\mathbb{Z}/n\mathbb{Z}[t]$ described above. Then we have that a set S is piecewise syndetic in $\mathbb{N}$ if and only if $\mathfrak{E}_n(S)$ is piecewise syndetic in $\mathbb{Z}/n\mathbb{Z}[t]$.*

Proof. This follows trivially from Property 1.11 and the definition of piecewise syndecity being the same in $\mathbb{N}$ and $\mathbb{Z}/n\mathbb{Z}[t]$.

Property 1.13 *Let $\mathcal{Q}$ be the family of piecewise syndetic sets. Then $\mathcal{Q}$ is upward closed and partition regular.*

Proof. Simply combine Property 1.8 and Definitions 1.9, 1.10 and 1.12.

Now that we have assembled our arsenal, it is time to go on the attack.

Theorem 3: Piecewise intersectivity

Let $A \subseteq \mathbb{Z}/n\mathbb{Z}[t]$ be piecewise syndetic. Then there exists a syndetic set L such that for any finite, non-empty $F \subseteq L$ the intersection

$$\bigcap_{q \in F} (A - n)$$

is piecewise syndetic.

Proof. We know that $A = S \cap T$ with S syndetic and T thick. Since S is syndetic, it also has a signature - say k - where $S + \mathbb{Z}/n\mathbb{Z}[t]_{\leq k} = \mathbb{Z}/n\mathbb{Z}[t]$. As mentioned before, T contains equivalency classes of arbitrarily large signature, and so, passing to a subset of T if necessary, we may write

$$T = T_k \cup T_{k+1} \cup T_{k+2} \cup \dots$$

where T_i is an equivalency class of signature i . We will construct a syndetic set L of signature k satisfying the above property. To do so, we begin by constructing an well-ordering on all the equivalency classes in $\mathbb{Z}/n\mathbb{Z}[t]/\mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$. The ordering is simply to consider the "smallest" element in the class in the ordering induced by the mapping $\mathfrak{E}_n^{-1}$ to $\mathbb{N}$. We can then just compare all the smallest elements.

Recall that $[\]_{\sim}$ denotes an equivalency class. We will now proceed by induction, by proving that there is a choice of $q_1, \dots, q_M$ from $[E_1]_{\sim}, \dots, [E_M]_{\sim}$ such that $\bigcap_{i=1}^M (A - q_i)$ is piecewise syndetic for all M . We then conclude by taking $L = \{q_1, \dots\}$ as we simply have to show the latter holds for arbitrarily large M for it to hold for finite subsets of L thus defined.

Suppose that some set $q_1, \dots, q_M$ (possibly with $M = 0$ - the empty set is the base case) already been found, and so by inductive hypothesis $A_M = \bigcap_{i=1}^M (A - q_i)$ is piecewise syndetic (in the base case, just take $A_0 = A$). Our goal is to find a class representative q_{M+1} . Choosing Q as an arbitrary element of $[E_{m+1}]$ we have that $[E_{m+1}] = \{Q - f \mid f \in \mathbb{Z}/n\mathbb{Z}[t]_{\leq k}\}$. As $S + \mathbb{Z}/n\mathbb{Z}[t]_{\leq k} = \mathbb{Z}/n\mathbb{Z}[t]$, we can also write that $\bigcap_{f \in \mathbb{Z}/n\mathbb{Z}[t]_{\leq k}} (S - f) = \mathbb{Z}/n\mathbb{Z}[t]$, recalling that as we are in a field, addition by an element in $\mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$ corresponds to subtraction by another element in $\mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$. This means that $\bigcap_{f \in [E_{m+1}]} (S - f) = \mathbb{Z}/n\mathbb{Z}[t] - Q = \mathbb{Z}/n\mathbb{Z}[t]$, as a shift by Q is injective. We can therefore write $A_M = \bigcup_{f \in \mathbb{Z}/n\mathbb{Z}[t]_{\leq k}} A \cap (S - Q - f)$ and this provides us with a finite partitioning of A_M ; therefore, by Property 1.14, we have $g \in \mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$ such that $A_M \cup (S - Q - g)$ is piecewise syndetic. Furthermore, we have $T = T_k \cup T_{k+1} \cup \dots$ and clearly, $(T_k - Q - g) = T_k$; $(T_{k+1} - Q - g) = T_{k+1}, \dots$ as shifting by a polynomial that is smaller than the signature is just a bijection of an equivalency class. We therefore have that $(T - Q - g) = T$ and so, taking $q_{M+1} := Q + g$,

$$A_M \cap (A - q_{M+1}) = A_M \cap (S - q_{M+1}) \cap (T - q_{M+1}) = A_M \cap (S - q_{M+1})$$

which is piecewise syndetic as desired.

Now, let us look at what results allow us to approach Van der Waerden's theorem.

1.4 Van der Waerden's Theorem in $\mathbb{Z}/n\mathbb{Z}[t]$

Before proving Van der Waerden's theorem itself, we prove a preparatory lemma.

Lemma 4: Fixed-value Van der Waerden

The following are equivalent:

1. For any rooted Φ_k and a finite colouring of $\mathbb{Z}/n\mathbb{Z}[t]$ there exists a monochromatic Φ_k -AP.
2. For any rooted Φ_k and $m \in \mathbb{N}$ there exists $N(m, k)$ such that if the set $\mathbb{Z}/n\mathbb{Z}[t]_{\leq N(m, k)}$ is coloured in m colours, there is a monochromatic Φ_k -AP in $\mathbb{Z}/n\mathbb{Z}[t]_{\leq N(m, k)}$.
3. For any rooted $\Phi_k \subset \text{End}(\mathbb{Z}/n\mathbb{Z}[t])$, any syndetic set $S \in \mathbb{Z}/n\mathbb{Z}[t]$ contains a Φ_k -AP.
4. For any piecewise syndetic $A \in \mathbb{Z}/n\mathbb{Z}[t]$ and Φ_k there exists $d \in \mathbb{Z}/n\mathbb{Z}[t]$ and a piecewise syndetic set B such that for all $b \in B$ we have $\{b + \Phi(d)\} \subset A$.

Proof that 1. $\implies$ 2. We will prove the contrapositive, by using the existence of a counterexample in 2 to create a counterexample for 1. Suppose now that there exists m such that there is no corresponding $N(m, k)$; or in other words for all $N \in \mathbb{N}$ there exists a colouring of $\mathbb{Z}/n\mathbb{Z}[t]_{\leq N}$ with no monochromatic ϕ_k -AP. Let χ_1 be such a colouring for $\mathbb{Z}/n\mathbb{Z}[t]_{\leq 1}$, χ_2 be such a colouring for $\mathbb{Z}/n\mathbb{Z}[t]_{\leq 2}$ and so forth. Now extend these colourings to all of $\mathbb{Z}/n\mathbb{Z}[t]$ by assigning an unused colour - say black - to all the elements of $\mathbb{Z}/n\mathbb{Z}[t] \setminus \mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$ in the colouring χ_k .

We can reprise our ordering of $\mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$ from the previous theorem, as it easily gives us a valid ordering of all of $\mathbb{Z}/n\mathbb{Z}[t]$ with the same principle (comparing from the largest coefficient downwards). As before, we use the ordering to give us an enumeration $q_1, q_2, \dots$ of all the elements of $\mathbb{Z}/n\mathbb{Z}[t]$.

Now, consider the various attributed colours of q_1 in $\chi_1(q_1), \chi_2(q_2), \dots$: this is an infinite sequence whose elements come from a finite set (the m colours and black) and so there is some colour that appears infinitely many times. Furthermore, this colour is not black, as there are only finitely many χ_k that assign the colour black to any given polynomial - once $k > \deg(q_i)$, the polynomial has been assigned one of the m original colours. We refer to this colour as $\chi(q_1)$; now consider the infinite subset $N_1 \subseteq \mathbb{N} := \{n \in \mathbb{N} \mid \chi_n(q_1) = \chi(q_1)\}$ of all the indices where q_1 was assigned this colour. We may now consider the colours attributed to q_2 by the $\chi_m, m \in N_1$; again, this gives an infinite sequence whose elements come from a finite set allowing us to attribute $\chi(q_2)$ in the same manner, yielding N_2 and allowing us to proceed with $\chi(q_3), \chi(q_4), \dots$ which generates a colouring χ of all of $\mathbb{Z}/n\mathbb{Z}[t]$. But this colouring clearly contains no monochromatic ϕ_k -AP, as if we suppose it did then we have $q_{i_1}, q_{i_2}, \dots, q_{i_k}$ a monochromatic ϕ_k -AP, implying for any $n \in N_{i_k}$, $\chi_n(q_{i_1}) = \dots = \chi_n(q_{i_k})$, contradicting the assumption we made at the beginning.

Proof that 2. $\implies$ 1. Trivial.

Proof that 1. $\implies$ 3. Let $S \subseteq \mathbb{Z}/n\mathbb{Z}[t]$ be syndetic with signature k ; we therefore have that $\mathbb{Z}/n\mathbb{Z}[t] = \bigcup_{f \in \mathbb{Z}/n\mathbb{Z}[t]_{\leq k}} (S - f)$. As this yields a colouring of $\mathbb{Z}/n\mathbb{Z}[t]$ with p^k colours (just identify every element to any one of the sets $S - f_i$ in which it belongs; there are p^k

elements in $\mathbb{Z}/n\mathbb{Z}[t]$, by 1., there is a monochromatic ϕ_k -AP in some $(S - g)$. This yields a ϕ_k -AP in S by adding g , as ϕ_k -APs are shift invariant by changing the value of the root.

Proof that 3. $\implies$ 4. Let $A \subseteq \mathbb{Z}/n\mathbb{Z}[t]$ be piecewise syndetic, and $\Phi'_k = \{\phi_1, \dots, \phi_k\} \subset \text{End}(\mathbb{Z}/n\mathbb{Z}[t])$ be a set as described in the statement of 4. Let $\Phi_k := \{0, \phi_2 - \phi_1, \dots, \phi_k - \phi_1\}$; note that the difference of two endomorphisms is also an endomorphism. Using Theorem 3 we can find a syndetic L such that for any finite subset $F \subseteq L$, we have that

$$\bigcap_{q \in F} (A - q)$$

is piecewise syndetic. By assumption, L contains a ϕ_k -AP; in other words, there exists a root a and difference d where $a, a + \phi_2(d) - \phi_1(d), \dots, a + \phi_k(d) - \phi_1(d)$ are all in L . Combining these facts, we have

$$\bigcap_{i=1}^k (A - a - \phi_i(d) + \phi_1(d))$$

is piecewise syndetic. Shifting this set by $-\phi_1(d)$, we have that

$$B := \bigcap_{i=1}^k (A - a - \phi_i(d))$$

is also piecewise syndetic. It follows that for any $b \in B$, we have $\{b + \phi_1(d), \dots, b + \phi_k(d)\} \subset A$, due to the above intersection.

Proof that 4. $\implies$ 1. If $\mathbb{Z}/n\mathbb{Z}[t]$ is coloured with finitely many colours then by Property 1.14, there is a monochromatic piecewise syndetic set. Therefore, by the assumption made in 4., we have a monochromatic ϕ_k -AP as desired.

We now want to prove Van de Waerden's theorem via induction, by increasing the size of the arithmetic progression. To formalise this notion of increasing the size, we state and prove one last lemma.

Lemma 5: Polynomial Color-focusing

Let $k \in \mathbb{N}$ and suppose we know for any given rooted Φ_k , a finite colouring of $\mathbb{Z}/n\mathbb{Z}[t]$ yields a monochromatic Φ_k -AP. Then for $\Phi'_k = \{\phi_1, \phi_2, \dots, \phi_k\}$, and any finite colouring of $\mathbb{Z}/n\mathbb{Z}[t]$, and any $r \in \mathbb{N}$, there exist monochromatic piecewise syndetic sets $A_0, A_1 \dots A_r \subseteq \mathbb{Z}/n\mathbb{Z}[t]$ such that for all $0 \leq i < j \leq r$ there exists a difference u where

$$\{a + \Phi'(u) \subseteq A_i.$$

Proof. Let us induct on r . Property 1.14 tells us that there exists a monochromatic piecewise syndetic set $A_0 \in \mathbb{Z}/n\mathbb{Z}[t]$. Suppose now we have found sets up to A_{r-1} . As we are assuming Part 1 of Lemma 4 holds for r , we also have that Part 4 of Lemma 4 holds and there exists a piecewise syndetic set B and a difference d such that for all $b \in B$,

$$\{b + \phi_1(d), \dots, b + \phi_k(d)\} \subset A_{r-1}.$$

Consider our finite colouring of $\mathbb{Z}/n\mathbb{Z}[t]$ restricted to B ; by Property 1.14 we have a monochromatic piecewise syndetic set $A_r \subseteq B$. By construction, for all $a \in A_r$,

$$\{a + \phi_1(d), \dots, a + \phi_k(d)\} \subset A_{r-1}.$$

Now, let $0 \leq i < j \leq r$. If $j < r$ then the existence of a difference u as described follows by the induction hypothesis so we may assume $j = r$. Now we use the induction hypothesis to instead find some u' such that for all $a \in A_{r-1}$,

$$\{a + \phi_1(u'), \dots, a + \phi_k(u')\} \subset A_i.$$

Now it suffices to take $u = u' + d$, with the d previously found, to obtain that for all $a \in A_r$,

$$\{a + \phi_1(u), \dots, a + \phi_k(u)\} \subset A_i$$

as by the definition of a homomorphism, $\phi_k(u' + d) = \phi_k(u') + \phi_k(d)$.

Theorem 6: Van der Waerden's Theorem in $\mathbb{Z}/n\mathbb{Z}[t]$

For any $k \geq 2 \in \mathbb{N}$, any rooted $\Phi_k \subset \text{End}(\mathbb{Z}/n\mathbb{Z}[t])$, and any finite colouring of $\mathbb{Z}/n\mathbb{Z}[t]$, there exists a monochromatic rooted Φ_k -AP.

Proof. We proceed by induction on k . The case $k = 2$ is the base case; we wish to show for an endomorphism ϕ there exists a, b such that $a, a + \phi(b)$ have the same colour. Firstly, consider $\text{Im}(\phi) \subseteq \mathbb{Z}/n\mathbb{Z}[t]$; if this set has finite cardinality then in particular $\exists c \neq d$ such that $\phi(c) = \phi(d) \implies \phi(c - d) = 0$, as since the domain has infinite cardinality but the image finite cardinality, the function cannot be injective. It therefore follows that the image has infinite cardinality, and by considering the restriction of our finite colouring to the image we know $\exists c \neq d$ such that $\phi(c), \phi(d)$ have the same colour, meaning $a := \phi(c), b := d - c$ yields a desired pair.

Suppose now that we have proved Van der Waerden up to some $k \geq 2$. Let $\Phi_{k+1} = \{\mathbf{0}, \phi_1, \dots, \phi_k\}$; if we use m colours in our colouring, then apply Lemma 2 to find monochromatic piecewise syndetic sets $A_0, A_1, \dots, A_m$ such that for all $0 \leq i < j \leq m$, we have $u \in \mathbb{Z}/n\mathbb{Z}[t]$ with

$$\{a + \phi_1(u), \dots, a + \phi_k(u) : a \in A_j\} \subseteq A_i.$$

By the pigeonhole principle, two of these sets must have the same colour. Consider these two sets A_i and A_j ; by taking any $a \in A_i$ we have that

$$\{a, a + \phi_1(u), \dots, a + \phi_k(u)\}$$

is a monochromatic Φ_{k+1} -AP.

Of course, by ignoring the first term of a rooted Φ_k -AP, we obtain a Φ_{k-1} -AP, so we can drop the necessary of being rooted. Furthermore, as we can find arbitrarily long arithmetic progressions, it stands to reason that we can also find arbitrarily many arithmetic sequences, just by treating each subsequence of a given length separately. Finally, we note that a sufficient condition describing a set for it to have this arithmetic progressions is that it be piecewise syndetic.

Monochromatic $p, p + q, pq$

Now, we take Moreira's proof of the existence of a monochromatic triplet $\{x, xy, x + y\}$ and reproduce it in our context.

2.1 Monochromatic $p, p + q$ and pq

We begin by converting our version of Van der Waerden's theorem to a slightly weaker result in a form that is more to our convenience.

Lemma 7: Alternative Van der Waerden

Let $E \in \mathbb{Z}/n\mathbb{Z}[t]$ be piecewise syndetic and let $F \in \mathbb{Z}/n\mathbb{Z}[t]$ be finite. Then there exist infinitely many $p \in \mathbb{Z}/n\mathbb{Z}[t]$ such that the intersection

$$E \cap \bigcap_{q \in F} (E - pq)$$

is piecewise syndetic.

Proof. Firstly, note that the set $S_p := 0 \cup \{pq \mid q \in F\}$ is simply equal to $\Phi(p) = \{\phi_0(p) = 0, \phi_1(p) = q_1 \cdot p, \dots\}$, a set of images of p via homomorphisms, as multiplication by a polynomial is a linear endomorphism. It follows that

$$a \in E \cap \bigcap_{q \in F} (E - pq) \iff a + \Phi(q) \in E$$

However, by Van der Waerden, we know there are infinitely many such arithmetic progressions, and so the set of roots of these arithmetic progressions - the union of all viable a - is the infinite set we are looking for. In particular, the set of viable a is a piecewise syndetic set.

Theorem 8: Moreira's result in $\mathbb{Z}/p\mathbb{Z}[t]$

Consider a finite colouring of $\mathbb{Z}/n\mathbb{Z}[t]$. Then there exist $p, q \in \mathbb{Z}/n\mathbb{Z}[t]$ such that the set $\{p, pq, p + q\}$ is monochromatic.

Proof. Let the colours used be enumerated as $1, 2, \dots, k$, and the respective sets as $C_1, C_2, \dots, C_k$. We will recursively construct four sequences:

- a sequence $(p_i)_{i \geq 1}$ of polynomials such that $\deg(p_i) < \deg(p_{i+1})$;

- two sequences $(B_i)_{i \geq 0}$ and $(D_i)_{i \geq 1}$ of piecewise syndetic subsets of $\mathbb{Z}/p\mathbb{Z}[t]$;
- a sequence $(t_i)_{i \geq 0}$ of colours

satisfying $B_i \subset C_{t_i}$.

We initially take t_0 such that C_{t_0} is piecewise syndetic; let $B_0 := C_{t_0}$. This are our base values to build upon. Suppose now we have $n \geq 0$ and we have already defined all the sequences up to $n-1$. We apply Lemma 7 in order to find $p_i \in \mathbb{Z}/p\mathbb{Z}[t]$, $\deg(p_i) > \deg(p_{i-1})$ such that for $F := \{1, p_{i-1}^2, p_{i-2}^2 p_{i-1}^2, \dots, p_1^2 \cdots p_{i-1}^2\}$ and $E := B_{i-1}$ the result holds; in other words,

$$B_{i-1} \bigcup_{j=1}^i (B_{i-1} - p_j^2 \cdots p_{i-1}^2 p_i)$$

is piecewise syndetic. Due to the set of viable p_i being piecewise syndetic, we can in particular choose one such that their product does not give 0, as for any nontrivial homomorphism, a piecewise syndetic set without the kernel of the homomorphism is still piecewise syndetic. It follows that $p_i D_i \cap C_{t_i}$ is piecewise syndetic for some t_i , by Property 1.14. This concludes the recursive generation of the sequences.

Furthermore, $y_i D_i$ is also piecewise syndetic - to see why, we can consider the alternative characterization of piecewise syndetic sets as being those who, via a finite union of shifts, yield a thick set - in other words, their sumset with some $\mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$ yields a thick set. If by assumption $D_i + \mathbb{Z}/n\mathbb{Z}[t]_{\leq k}$ is thick, then $y_i D_i + \mathbb{Z}/n\mathbb{Z}[t]_{\leq k + \deg(y_i)}$ yields the same thick set up to a shift, which of course does not affect thickness.

By the above, we have $B_i \subset p_i B_{i-1}$; iterating this, we obtain $B_i \subset p_i p_{i-1} \cdots p_{j+1}$ for any $j < i$.

As the sequence $(t_i)_{i \geq 0}$ contains only finitely many colours, we must have $t_i = t_j$ at some point (indeed, within the first $k+1$ terms). Consider $z \in B_i$, $y := p_i p_{i-1} \cdots p_{j+1}$ and $x = \frac{z}{y}$, which we know is polynomial as all the elements of B_i are divisible by $p_i p_{i-1} \cdots p_1$. By definition, $xy = z \in C_{t_i}$ and as $B_i = y B_j$, $x \in B_j \implies x \in C_{t_j} = C_{t_i}$. We conclude by noting that

$$\begin{aligned} y(x+y) &= z + y^2 \in B_i + y^2 \subset D_i + y^2 \\ &\subset p_i(B_{i-1} - p_i p_{i-1}^2 \cdots p_{j+1}^2) + y^2 \\ &\subset p_i(p_i p_{i-1} \cdots p_{j+1} B_j - p_i p_{i-1}^2 \cdots p_{j+1}^2) + y^2 \\ &= y B_j - y^2 + y^2 \subseteq y B_j \end{aligned}$$

implying $x+y \in B_j \subseteq C_{t_j} = C_{t_i}$ and yielding the desired conclusion.

A natural question to ask is whether this result can be extended to $\mathbb{Z}/n\mathbb{Z}$. However, the zero-divisors issue is difficult to surmount. For example, suppose that we are working with $n = 8$ and have already found $p_1 = 2$, and that B_1 is the set of all polynomials whose coefficients are even.. This is piecewise syndetic, as it is the intersection of a syndetic set (all polynomials whose constant term is even) and a thick set (all polynomials whose non-constant terms are even).

2.2 Polynomials XOR integers?

There is a clear correspondence between the polynomials in $\mathbb{Z}/n\mathbb{Z}[t]$ and the numbers in base n ; one that is obtained by $p \in \mathbb{Z}/n\mathbb{Z}[t] \rightarrow p(n) \in \mathbb{N}$, the application that essentially extracts the coefficients from a polynomial. For instance, for $n = 10$ we have that $p = 4t^2 + 6t + 7$ is associated to the number 467. In particular, this is the unique bijection $\mathfrak{E}$ from $\mathbb{Z}/n\mathbb{Z}[t]$ to $\mathbb{N}$ that preserves the total order defined previously.

It is now natural to ask oneself how addition and multiplication in these two contexts correspond. The answer is that they are very similar, but in the polynomial case there is no notion of "carrying over" and instead when coefficients exceed n , they just roll over back to 0 without contributing anything to the next higher coefficient as is the case with the integers. For instance, with $n = 10$,

$$\begin{aligned} 526 + 704 &= 1230. \\ (5t^2 + 2t + 6) + (7t^2 + 4) &= 2t^2 + 2t. \end{aligned}$$

The most intuitive way to merge both operations is to define a new operation on the integers - $\oplus$ - which adds modulo each digit and corresponds to the polynomial version of addition. In base 2, for instance, it corresponds exactly to the bitwise XOR operation. Moreira's original result proved the existence of a monochromatic triplet $\{x, x + y, x \times y\}$; Theorem 4 proves $\{x, x \oplus y, x \otimes y\}$. The question is now whether we can mix the two: might it be possible to find $\{x, x + y, x \oplus y\}$?⁴

The first thing to decide is exactly how our correspondence will work - for example, if we were to begin by colouring the integers and then translate an integer into polynomials, the integer 16 corresponds to different polynomials based on which n we take - it is t^4 for $n = 2$, $t^2 + 2t + 1$ for $n = 3$, t^2 for $n = 4$ and so forth. To compute $a \oplus b$, one would turn the two into polynomials, compute the addition in $\mathbb{Z}/n\mathbb{Z}[t]$ and then turn the result back into an integer. Of course, thus defined, the $\oplus$ and $\otimes$ operators depend on n , and so we would denote them as $\oplus_n$ and $\otimes_n$ respectively. Alternatively, if we were to begin by colouring the polynomials rather than integers, we would consider the polynomial $t^2 + 6$ to represent different integers based on which base we take it in - $n = 2$ yields 10, $n = 3$ yields 15 and so forth. However, if we proceed like this then we have the exact same contextualisation, because we are just traversing the bijection in the opposite direction.

A truly different approach is to instead begin at the set of all finite sequences of positive integers, which we will denote $\omega^{<\omega}$ as is the set-theoretic standard. Each such sequence corresponds to both a polynomial and an integer, for any choice of basis larger than the largest term in the sequence. For instance, the sequence $(6, 2, 7)$ can be read as the integers $627_8 (= 407_{10})$, $627_9 (= 511_{10})$, 627_{10} and so forth depending on what base we are writing the integers in. Similarly, for the polynomials, although we always consider

⁴We will not attempt to work with the multiplication operators, as they are far too intricate.

$6t^2+2t+7$, how addition works depends exactly on the value of n in $\mathbb{Z}/n\mathbb{Z}$. In this case, both $+$, $\times$ and $\oplus, \otimes$ depend on the respective bases, and so we denote them $+_m, \times_m, \oplus_n, \otimes_n$. Now, we can first choose m and n and then restrict ourselves to those sequences whose largest term is strictly smaller than the minimum of the two. Note that this is not a generalisation of the former problem, as fixing $m = 10$ might yield the standard integers, but the application from integers to polynomials is no longer the same - indeed, we do not even have a bijection anymore unless $n = 10$ as well.

We therefore have two different problems to solve now, the existence of a monochromatic $\{x, x + y, x \oplus_n y\} \subset \mathbb{N}$ and the same for $\{u, u +_m v, u \oplus_n v\} \subset \omega^{<\omega}$, with $m, n \geq 2$. We will show that both can be found - along with y or v in the same colour - using the same ingenious method.

2.3 Game, set and match

A natural way to simplify our problem is to ask ourselves when $x + y = x \oplus_n y$, or when $u +_m v = u \oplus_n v$, as restricting ourselves to these cases means we only need to find a triplet (with the caveat that we are working on a strictly stronger result and a failure to find such a triplet does not mean that the more general quadruplet does not exist).

In the first case, when does $\oplus_n$ correspond to integer addition? Differences arise when two coefficients in the same position sum to more than n ; it is here that we would "carry forward" in the integers but loop back in the polynomials. The simplest way to ensure equality is to therefore choose x and y such that their coefficient-wise sums are less than n - for example, $136 + 102 = 136 \oplus_3 102$ as in base 3 they are equal to 12001_3 and 10210_3 respectively, and there is no overflow. Similarly, in the second case if we just ensure the same condition holds on the strings u and v , we know $u +_m v = u \oplus_n v$.

If the result were to hold for all m, n as desired, it would in particular hold for $m = n = 2$, in which case we are restricted to taking two binary polynomials/sequences which never have a 1 in the same position. Naturally, if the result holds for all colourings of such polynomials/sequences then it holds for *any* choice of m, n , because we can always restrict ourselves to those that only contain 0s and 1s. In other words, if we were doing the first problem, and we knew the aforementioned result held, we could restrict ourselves to those integers containing only 0s and 1s in their base- n expansion. For the second problem we would proceed likewise, only considering strings containing 0s and 1s.

Now, every such integer or string can be described by the (finite) set of coefficients where it has a 1. In other words, we are colouring all the finite subsets of $\mathbb{N}$ (which we refer to as $\mathcal{P}_{<\omega}(\mathbb{N})$) and searching for disjoint A, B such that $A, B, A \cup B$ are monochromatic. We have therefore linked our number-theoretic problem to one in set theory, which we will now prove holds.

Theorem 9: Monochromatic sets

For any finite colouring of $\mathcal{P}_{<\omega}(\mathbb{N})$, we can find disjoint sets $A, B \in \mathcal{P}_{<\omega}(\mathbb{N})$ such that $\{A, B, A \cup B\}$ are monochromatic.

Proof. We use the notation k -sets to refer to all the sets of cardinality k .

Consider all the 1-sets, or singletons. By the infinite pigeonhole principle, there must be an infinite set of singletons that have the same colour. Call this set N_1 and the corresponding colour c_1 .

Now, consider all the 2-sets of N_1 , usually denoted by $N_1^{(2)}$. By Ramsey's theorem on 2-sets, we have an infinite subset N_2 such that $N_2^{(2)}$ is monochromatic, with a colour c_2 .

By Ramsey's theorem on 3-sets, we can find a monochromatic $N_3^{(3)}$, and proceed to find $N_3 \supset N_4 \supset N_5 \supset \dots$ in this manner using Ramsey on k -sets. Now, the finite version of Schur's theorem tells us for any r there is an integer $M(r)$ such that any finite colouring of $1, 2, \dots, M(r)$ yields a monochromatic triplet $\{x, y, x + y\}$. Therefore, if we now consider the colours $c_1, c_2, \dots, c_{M(r)}$ we have a triplet $c_i = c_j = c_{i+j}$. Consider any $i + j$ -set C in N_{i+j} and partition it into two arbitrary disjoint subsets A and B of cardinality i and j respectively. By construction, $A \subset N_{i+j} \subset N_i$ and $B \subset N_{i+j} \subset N_j$. It follows that A, B and $C = A \cup B$ are monochromatic.

Of course, the assumption of disjointness can be dropped to give a weaker but more general statement. Also, it follows that there are infinitely many such pairs. Our two earlier conjectures are hence proven.

Corollary 10: Integer-based mixed quadruplets

For any finite colouring of $\mathbb{N}$, and any n we can find (infinitely many) quadruplets $\{x, y, x + y, x \oplus_n y\} \subset \mathbb{N}$ that are monochromatic, where

$$x \oplus_n y = \sum_{i=0}^{\lfloor \log_n(x) \rfloor} n^i \left[\left(\left\lfloor \frac{x}{n^i} \right\rfloor + \left\lfloor \frac{y}{n^i} \right\rfloor \right) \bmod n \right].$$

Corollary 11: String-based mixed quadruplets

For any finite colouring of $\omega^{<\omega}$, and any n, m we can find (infinitely many) quadruplets $\{u, v, u +_m v, u \oplus_n v\} \subset \omega^{<\omega}$ that are monochromatic, where (denoting ϕ_m, ϕ_n and ϕ_m^{-1}, ϕ_n^{-1} as the transformations to $\mathbb{N}$ and vice versa where a string is read in base m or n , and an integer in base m or n is read as a string)

$$\begin{aligned} u +_m v &= \phi_m^{-1}(\phi_m(u) + \phi_m(v)). \\ u \oplus_n v &= \phi_n^{-1}(\phi_n(u) \oplus_n \phi_n(v)). \end{aligned}$$

2.4 To infinity and beyond

One final question we will now tackle is the following generalisation of Theorem 9: can we find an arbitrarily large family of sets that is monochromatic under unions? And can we find an infinite family of sets where this property still holds?

Since the latter is the stronger statement, we will attack it first. A natural starting point is Hindman's theorem, which guarantees the existence of monochromatic IP sets⁵. The issue with trying to apply it naively to our proof of Theorem 9 is that the heuristic of creating a sequence of nested sets does not extend to infinity: it is possible the intersection of all the sets in our sequence is empty, meaning that whilst we can create arbitrarily large sets where the union is monochromatic, we cannot find an infinite set in this manner.

To proceed, we will need to apply a clever use of binary representation.

Theorem 12: Monochromatic infinite sets

For any finite colouring of $\mathcal{P}_{<\omega}(\mathbb{N})$, we can find an infinite family of disjoint sets $S \subseteq \mathcal{P}_{<\omega}(\mathbb{N})$ such that for any finite subset F of S , the union

$$\bigcup_{s \in F} s$$

has the same colour regardless of the choice of F .

Proof. Consider a finite colouring of $\mathcal{P}_{<\omega}(\mathbb{N})$. Now, we transform this into a colouring of $\mathbb{N}$; let $\mathfrak{F} : \mathcal{P}_{<\omega}(\mathbb{N}) \rightarrow \mathbb{N}$ be defined by

$$\mathfrak{F}(S) = \sum_{i \in S} 2^{i-1}$$

This is clearly a bijection by the uniqueness of the binary representation of the integers, and therefore endows $\mathbb{N}$ with a colouring. By Hindman's theorem, under this colouring of $\mathbb{N}$ we have an infinite set $S = s_1 < s_2 < s_3 < \dots$ such that the finite sums of S are monochromatic. Now, we construct a second set T . Let $t_1 = s_1$. Suppose we have found the first k elements of T . Pick $N_k \in \mathbb{N}$ such that $2^{N_k} > t_k$. As S is an infinite set, it contains infinitely many integers that are $\equiv m \pmod{2^{N_k}}$ for some m . Pick the first 2^{N_k} of these and sum them to obtain t_{k+1} - although this is not necessarily an element of S , it is a finite sum of elements of S and therefore replacing the elements we used to create it with this new number does not change the fact that the finite sums of S are monochromatic, since doing so only reduces the set of finite sums. Naturally, by construction, $2^{N_k} | t_{k+1}$, and so t_{k+1} contains no binary digits in common with $t_k, t_{k-1}, \dots, t_1$. We proceed in this manner to create an infinite set T . Now, consider any finite subset F of T . In particular, $\mathfrak{F}^{-1}(F)$ is a set of disjoint subsets of $\mathcal{P}_{<\omega}(\mathbb{N})$. Consequently,

$$\mathfrak{F}^{-1}\left(\bigcup_{s \in F} s\right) = \bigcup_{s \in F} \mathfrak{F}^{-1}(s)$$

and as the latter is monochromatic regardless of the choice of F by construction, so is the former. It follows that $\mathfrak{F}^{-1}(T)$ satisfies the desired properties.

⁵An IP set is a set containing all finite sums of an infinite set. More precisely, for any IP set T there is an infinite set of integers $S = s_1, s_2, \dots$ such that $\forall I \in \mathcal{P}_{<\omega}(\mathbb{N}), \sum_{i \in I} s_i \in T$.